

Host-Only Static IP Client-Server Network Lab

1. The Computers

Server (Hosts services)

IP: 192.168.100.10

Role: Runs Apache web server

Client (Tests and connects to server)

IP: 192.168.100.20

Role: Connects to server and verifies services

Network Type: Host-Only (Private internal LAN only)

Subnet: 192.168.100.0/24

Meaning: Both machines must start with 192.168.100 to communicate.

2. What I Did

Placed both VMs on Host-Only network.

Disabled DHCP and assigned static IP addresses using netplan.

Applied configuration with: `sudo netplan apply`

Verified IP assignment using: `ip a`

Tested connectivity using: `ping 192.168.100.10`

Installed Apache on server: `sudo apt install apache2`

Verified web service from client using: `curl http://192.168.100.10`

3. What It Means (Simplified)

Static IP = Manually assigned address.

DHCP = Automatic addressing (disabled for this lab).

/24 subnet = First three numbers must match.

Ping = Checks if the other computer is reachable.

Apache = Web server software.

Host-Only = Isolated internal network.

No gateway needed because both systems are on the same subnet.

4. What I Learned

How to manually configure network settings.

How subnetting controls communication.

How client-server communication works inside a LAN.

How services (like Apache) affect network accessibility.

How incorrect IP or subnet configuration breaks connectivity.